

SAINTGITS COLLEGE OF ENGINEERING

Department of Electronics and Communication Engineering

MATHEMATICS ASSIGNMENT – PHASE 1

1D Image Blurring and Restoration Using the Heat Equation

Topic 2

GROUP 4

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Contents

Abstract	2
1 Introduction	3
2 Problem Statement and Mathematical Model	4
2.1 Objective	5
3 Separation of Variables	6
4 Spatial Equation and Boundary Conditions	7
4.1 Case 1: $\lambda = 0$	7
4.2 Case 2: $\lambda < 0$	7
4.3 Case 3: $\lambda > 0$	7
5 Time-Dependent Solution	9
6 Calculation of the Fourier Coefficients	10
7 Final Fourier-Series Solution	11
7.1 Special Case: Edge at the Midpoint	11
8 Mathematical Proof of Frequency-Dependent Decay	11
8.1 Higher Modes Decay Faster	12
8.2 Rate of Decay	12
8.3 Decay Time Scale	13
9 Why the Step Edge Becomes Blurred	13
10 Physical and Image-Processing Interpretation	14
11 Theoretical View of Image Restoration	14
12 Verification of the Solution	16
13 Key Results	17
14 Conclusion	18
References	18

Abstract

Image blurring can be understood as a diffusion process in which sharp changes in image intensity gradually become smooth. In this work, a one-dimensional image signal is modeled using the heat equation,

$$\frac{\partial I}{\partial t} = \alpha \frac{\partial^2 I}{\partial x^2}, \quad (1)$$

where $I(x, t)$ represents the image intensity and α is the diffusion or blurring coefficient. The equation is solved using the method of separation of variables along with a Fourier sine series.

A sharp step edge is considered as the initial image signal. The mathematical solution shows that each frequency component decreases over time according to an exponential decay factor. Higher-frequency components decay faster than lower-frequency components. Since sharp edges contain many high-frequency components, their rapid reduction causes the edges to become smoother, resulting in image blur.

This study also explains the basic idea of image restoration. Although the original image can theoretically be recovered by reversing the diffusion process, high-frequency noise is also strongly amplified, making exact restoration difficult. Overall, the analysis provides a clear mathematical explanation of how diffusion causes image blurring and why restoring a blurred image is a challenging problem.

1 Introduction

Image blurring is a common phenomenon in image processing in which sharp edges and fine details gradually become smooth. A useful mathematical model for understanding this process is the one-dimensional heat equation,

$$\frac{\partial I(x, t)}{\partial t} = \alpha \frac{\partial^2 I(x, t)}{\partial x^2}, \quad \alpha > 0,$$

where $I(x, t)$ represents the image intensity at position x and time t , and α is the diffusion or blurring coefficient. Similar to the way heat spreads from one region to another, the intensity values in an image also spread to neighbouring regions during diffusion.

Sharp edges represent sudden changes in intensity and contain high-frequency components. As the diffusion process continues, these high-frequency components decrease more rapidly than low-frequency components. Consequently, sharp edges gradually lose their definition, fine details become less visible, and the image appears blurred.

In this work, a sharp step function is used to represent an image edge. The heat equation is solved using the method of separation of variables, and the resulting solution is expressed as a Fourier sine series. This approach makes it possible to study how individual frequency components change with time. The analysis also provides insight into image restoration, showing that although blurring can theoretically be reversed, the restoration process can strongly amplify noise. Thus, the study connects the mathematical theory of partial differential equations and Fourier analysis with the practical problems of image blurring and restoration.

2 Problem Statement and Mathematical Model

Image blurring can be understood as a diffusion process in which image intensity gradually spreads from one region to another. This process reduces sudden changes in intensity and causes sharp edges to become smoother. To study this mathematically, we consider a one-dimensional image signal defined over a finite interval $0 < x < L$. Let $I(x, t)$ represent the image intensity at position x and time t . The blurring process is modeled using the one-dimensional heat equation

$$\frac{\partial I(x, t)}{\partial t} = \alpha \frac{\partial^2 I(x, t)}{\partial x^2}, \quad \alpha > 0. \quad (2)$$

Here, α represents the diffusion or blurring coefficient, which controls the rate at which the intensity spreads. Homogeneous boundary conditions are imposed at both ends of the interval:

$$I(0, t) = 0, \quad I(L, t) = 0, \quad t > 0. \quad (3)$$

A sharp edge is represented using a step function as the initial condition. If the edge is located at $x = a$, where $0 < a < L$, then

$$I(x, 0) = I_0 H(x - a), \quad (4)$$

where I_0 is the intensity level. The main objective is to determine how this initially sharp intensity profile changes with time. By solving the heat equation using separation of variables and Fourier series, the decay of different spatial frequency components can be studied, helping to explain mathematically why sharp edges gradually become blurred.

2.1 Objective

The main objective of this work is to understand image blurring using the mathematical model provided by the one-dimensional heat equation. The study focuses on how an initially sharp image signal changes over time as diffusion takes place.

The specific objectives are:

1. To solve the one-dimensional heat equation using the method of separation of variables.
2. To determine the spatial eigenvalues and eigenfunctions that satisfy the given boundary conditions.
3. To represent a sharp image edge using a step function and find its Fourier sine series coefficients.
4. To obtain the complete mathematical expression for the image intensity as it changes with time.
5. To show that high-frequency components decay faster than low-frequency components and explain how this leads to image blurring.
6. To understand the basic mathematical idea behind image restoration and the difficulties involved in reversing the blurring process.

3 Separation of Variables

Separation of variables is a mathematical method used to solve partial differential equations by expressing the unknown function as a product of two or more functions, each depending on a single independent variable. For the heat equation, the image intensity $I(x, t)$ is assumed to have the form

$$I(x, t) = X(x)T(t), \quad (5)$$

where $X(x)$ describes the variation of intensity with position and $T(t)$ describes how the intensity changes with time. This assumption allows the original partial differential equation to be separated into two simpler ordinary differential equations, which can then be solved individually. Let the required MSV solution is of the form

$$I(x, t) = X(x)T(t). \quad (6)$$

Substituting into the heat equation gives

$$X(x)T'(t) = \alpha X''(x)T(t).$$

Dividing by $\alpha X(x)T(t)$ gives

$$\frac{T'(t)}{\alpha T(t)} = \frac{X''(x)}{X(x)}. \quad (7)$$

Since the left-hand side depends only on t and the right-hand side depends only on x , both must equal a constant. Choose

$$\frac{T'}{\alpha T} = \frac{X''}{X} = -\lambda. \quad (8)$$

Therefore,

$$X'' + \lambda X = 0, \quad (9)$$

and

$$T' + \alpha \lambda T = 0. \quad (10)$$

4 Spatial Equation and Boundary Conditions

The spatial equation is

$$X'' + \lambda X = 0.$$

4.1 Case 1: $\lambda = 0$

If $\lambda = 0$,

$$X'' = 0,$$

so

$$X(x) = C_1 + C_2 x.$$

Applying $X(0) = 0$ and $X(L) = 0$ gives only the trivial solution.

4.2 Case 2: $\lambda < 0$

Let

$$\lambda = -\mu^2, \quad \mu > 0.$$

Then

$$X'' - \mu^2 X = 0.$$

The solution is

$$X(x) = A \cosh(\mu x) + B \sinh(\mu x).$$

Applying the boundary conditions again produces only the trivial solution.

4.3 Case 3: $\lambda > 0$

Let

$$\lambda = \mu^2.$$

Then

$$X'' + \mu^2 X = 0,$$

with solution

$$X(x) = A \cos(\mu x) + B \sin(\mu x).$$

From $X(0) = 0$,

$$A = 0.$$

Therefore,

$$X(x) = B \sin(\mu x).$$

Using $X(L) = 0$,

$$\sin(\mu L) = 0.$$

Thus,

$$\mu L = n\pi, \quad n = 1, 2, 3, \dots$$

and hence

$$\mu_n = \frac{n\pi}{L}, \quad \lambda_n = \left(\frac{n\pi}{L}\right)^2. \quad (11)$$

The corresponding eigenfunctions are

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right). \quad (12)$$

5 Time-Dependent Solution

For each eigenvalue λ_n ,

$$T'_n + \alpha\lambda_n T_n = 0.$$

Therefore,

$$\frac{T'_n}{T_n} = -\alpha\lambda_n.$$

Integrating,

$$\ln |T_n| = -\alpha\lambda_n t + C.$$

Hence,

$$T_n(t) = C_n e^{-\alpha\lambda_n t}.$$

Using

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2,$$

we obtain

$$T_n(t) = C_n e^{-\alpha(n\pi/L)^2 t}. \quad (13)$$

Therefore,

$$I_n(x, t) = C_n \sin\left(\frac{n\pi x}{L}\right) e^{-\alpha(n\pi/L)^2 t}.$$

By superposition,

$$I(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-\alpha(n\pi/L)^2 t}. \quad (14)$$

6 Calculation of the Fourier Coefficients

At $t = 0$,

$$I(x, 0) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right).$$

The Fourier sine coefficient is

$$B_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx. \quad (15)$$

For the step function,

$$f(x) = I_0 H(x - a),$$

therefore,

$$B_n = \frac{2I_0}{L} \int_a^L \sin\left(\frac{n\pi x}{L}\right) dx.$$

Let

$$k_n = \frac{n\pi}{L}.$$

Then

$$B_n = \frac{2I_0}{L} \left[-\frac{\cos(k_n x)}{k_n} \right]_a^L.$$

Since

$$\cos(k_n L) = \cos(n\pi) = (-1)^n,$$

we obtain

$$B_n = \frac{2I_0}{n\pi} \left[\cos\left(\frac{n\pi a}{L}\right) - (-1)^n \right]. \quad (16)$$

7 Final Fourier-Series Solution

Substituting B_n into the general solution gives

$$I(x, t) = \frac{2I_0}{\pi} \sum_{n=1}^{\infty} \frac{\cos\left(\frac{n\pi a}{L}\right) - (-1)^n \sin\left(\frac{n\pi x}{L}\right)}{n} \exp\left[-\alpha \left(\frac{n\pi}{L}\right)^2 t\right]. \quad (17)$$

For $t > 0$, the exponential factor suppresses the oscillatory components and the sharp transition becomes smooth.

7.1 Special Case: Edge at the Midpoint

To better understand the Fourier-series solution, consider the special case in which the sharp edge is located at the midpoint of the interval. This means that the edge position is chosen as

$$a = \frac{L}{2}.$$

With this choice, the initial intensity changes at the centre of the interval. Substituting $a = L/2$ into the Fourier coefficient expression simplifies the mathematical result and helps illustrate how the position of an edge affects the frequency components present in the image signal.

8 Mathematical Proof of Frequency-Dependent Decay

The Fourier-series solution of the heat equation contains an exponential factor for each spatial frequency component. This factor determines how the amplitude of each mode changes with time. Since the rate of decay depends on the mode number n , different frequency components do not decrease at the same rate. In particular, higher-frequency modes decay more rapidly than lower-frequency modes. This section provides a mathematical proof of this behaviour and explains its importance in the formation of image blur. The attenuation factor for the n th spatial Fourier mode is

$$D_n(t) = \exp\left[-\alpha \left(\frac{n\pi}{L}\right)^2 t\right]. \quad (18)$$

8.1 Higher Modes Decay Faster

Each term in the Fourier-series solution represents a spatial frequency mode, identified by its mode number n . As the value of n increases, the corresponding mode represents a higher-frequency variation in the image signal. The exponential decay factor depends on n^2 , which means that higher modes lose their amplitude more quickly over time. By comparing two modes with different values of n , we can mathematically show that the higher-frequency mode always decays faster. Consider two modes n and m with $n > m$. For $t > 0$,

$$n^2 > m^2,$$

therefore,

$$D_n(t) < D_m(t).$$

Thus, higher-frequency modes decay faster.

8.2 Rate of Decay

The rate of decay describes how quickly the amplitude of each Fourier mode decreases with time. By differentiating the decay factor, we can determine how this rate depends on the mode number n . Differentiating,

$$\frac{dD_n}{dt} = -\alpha \left(\frac{n\pi}{L} \right)^2 D_n(t).$$

Therefore,

$$\left| \frac{D'_n(t)}{D_n(t)} \right| = \alpha \left(\frac{n\pi}{L} \right)^2.$$

The relative decay rate is proportional to n^2 .

8.3 Decay Time Scale

The decay time scale gives a simple way to measure how quickly a particular frequency component fades during the diffusion process. It depends on the mode number n and the diffusion coefficient α . By defining a characteristic time for each mode, we can clearly see that higher-frequency components have shorter decay times and therefore disappear more rapidly. Define the characteristic decay time τ_n by

$$\alpha \left(\frac{n\pi}{L} \right)^2 \tau_n = 1.$$

Hence,

$$\tau_n = \frac{L^2}{\alpha \pi^2 n^2}. \quad (19)$$

Therefore,

$$\tau_n \propto \frac{1}{n^2}.$$

9 Why the Step Edge Becomes Blurred

A step edge represents a sudden change in image intensity and therefore contains a wide range of spatial frequency components. To produce a sharp transition, high-frequency modes are especially important. During the diffusion process, these higher-frequency components decay much faster than the lower-frequency components. As a result, the sharp edge gradually loses its abrupt shape and becomes smoother, which appears as image blur. A sharp edge contains many spatial frequencies. High-frequency modes are necessary to represent an abrupt transition.

For $t > 0$, every Fourier coefficient is multiplied by

$$e^{-\alpha(n\pi/L)^2 t}.$$

Since this factor decreases rapidly as n increases, high-frequency components disappear first. Therefore,

diffusion $\implies$ high-frequency attenuation $\implies$ loss of sharpness $\implies$ blur.

10 Physical and Image-Processing Interpretation

The parameter α controls the strength of diffusion. A larger α causes faster smoothing, while increasing t also increases the amount of blur.

The n th spatial mode has angular spatial frequency

$$\omega_n = \frac{n\pi}{L}.$$

Thus,

$$D(\omega_n, t) = e^{-\alpha\omega_n^2 t}.$$

This has the qualitative form of a low-pass filter: low spatial frequencies are retained while high spatial frequencies are strongly attenuated.

11 Theoretical View of Image Restoration

Image restoration aims to recover the original sharp image from a blurred observation. In the heat equation model, blurring reduces the high-frequency components that are responsible for sharp edges and fine details. Therefore, restoration attempts to recover these lost components.

Figure 1 illustrates the difference between an original sharp image and its blurred version. The original image has clear edges and fine details, while the blurred image has smoother transitions due to the decay of high-frequency components.

Mathematically, if B_n represents the original Fourier coefficient, then after diffusion for time t , the observed coefficient becomes

$$B_n^{\text{obs}}(t) = B_n e^{-\alpha\left(\frac{n\pi}{L}\right)^2 t}. \quad (20)$$

To theoretically recover the original coefficient, the inverse process would require



(a) Original sharp image



(b) Blurred image

(a) Original sharp image vs Blurred image

Figure 1: Comparison between an original sharp image and a blurred image. Diffusion reduces high-frequency components, causing edges and fine details to become smoother.

$$B_n = B_n^{\text{obs}}(t)e^{+\alpha\left(\frac{n\pi}{L}\right)^2 t}. \quad (21)$$

However, this restoration process is unstable because the exponential factor also amplifies noise, especially for high-frequency components. The forward heat equation describes blurring. If B_n is the initial Fourier coefficient, then after time t ,

$$B_n^{\text{obs}}(t) = B_n e^{-\alpha(n\pi/L)^2 t}.$$

Formally,

$$B_n = B_n^{\text{obs}}(t)e^{+\alpha(n\pi/L)^2 t}.$$

However, this inverse operation is unstable because high-frequency errors are also amplified exponentially. Therefore, direct restoration is mathematically ill-conditioned and generally requires regularization.

12 Verification of the Solution

Let

$$I(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-\alpha(n\pi/L)^2 t}.$$

Differentiating with respect to t ,

$$\frac{\partial I}{\partial t} = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) \left[-\alpha \left(\frac{n\pi}{L}\right)^2\right] e^{-\alpha(n\pi/L)^2 t}.$$

Differentiating twice with respect to x ,

$$\frac{\partial^2 I}{\partial x^2} = \sum_{n=1}^{\infty} B_n \left[-\left(\frac{n\pi}{L}\right)^2\right] \sin\left(\frac{n\pi x}{L}\right) e^{-\alpha(n\pi/L)^2 t}.$$

Thus,

$$\frac{\partial I}{\partial t} = \alpha \frac{\partial^2 I}{\partial x^2}.$$

Also,

$$I(0, t) = 0, \quad I(L, t) = 0,$$

so the boundary conditions are satisfied.

13 Key Results

Quantity	Result
Heat equation	$I_t = \alpha I_{xx}$
Boundary conditions	$I(0, t) = I(L, t) = 0$
Eigenvalues	$\lambda_n = \left(\frac{n\pi}{L}\right)^2$
Eigenfunctions	$X_n(x) = \sin\left(\frac{n\pi x}{L}\right)$
Time factor	$e^{-\alpha(n\pi/L)^2 t}$
Step input	$I(x, 0) = I_0 H(x - a)$
Fourier coefficient	$B_n = \frac{2I_0}{n\pi^2} \left[\cos\left(\frac{n\pi a}{L}\right) - (-1)^n \right]$
Decay time	$\tau_n = \frac{L^2}{\alpha\pi^2 n^2}$
High-frequency behavior	$D_n(t) \rightarrow 0 \quad as \quad n \rightarrow \infty$

14 Conclusion

This work presented a mathematical study of one-dimensional image blurring and restoration using the heat equation. The heat equation describes how image intensity changes over time due to diffusion. As diffusion takes place, intensity values spread to neighbouring regions, causing sharp changes and edges to become smoother.

The solution was obtained using the method of separation of variables. The image intensity was expressed as a product of spatial and time-dependent functions, which transformed the original partial differential equation into simpler ordinary differential equations. Applying the boundary conditions produced a set of eigenvalues and sine eigenfunctions, leading to a Fourier sine series representation of the solution.

A step function was used as the initial condition to represent a sharp edge. The analysis showed that each Fourier mode is multiplied by the exponential decay factor

$$e^{-\alpha\left(\frac{n\pi}{L}\right)^2 t}.$$

Since the decay depends on n^2 , higher-frequency modes decrease much faster than lower-frequency modes. As these high-frequency components are responsible for sharp edges and fine details, their rapid decay causes the image to become blurred.

The study also explained the challenge of image restoration. Although the blurring process can theoretically be reversed, high-frequency noise is amplified during restoration, making the inverse problem unstable. Overall, this analysis demonstrates the important connection between the heat equation, Fourier analysis, image blurring, and image restoration.

References

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